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Digital Product Passport: A Review and Research Agenda

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Abstract

Digital product passports (DPP) are at the forefront of circular economy initiatives and green logistics. They can be defined as unique digital documents that memorize and disclose product sustainability information throughout the lifecycle. However, DPP challenges are substantial, and research is still in its early stages and fragmented in this vibrant field that promises to change product information flows at an international scale. This paper presents a concept-centric systematic literature review and a bibliometric analysis of 40 publications published since 2021. The main objective is to understand the research landscape of DPP and identify guidelines for DPP structure, technology, requirements, and challenges for DPP adoption. A research agenda for DPPs is also presented. For academics, this review highlights the most recent advances in DPP research and unsolved questions. Practitioners may find our review helpful in identifying information requirements, preparing their digital infrastructure for DPP creation, and anticipating their role in DPP construction with particularities for each supply chain.

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Keywords: Digital product passport; Circular economy; Supply chain; Logistics; Systematic literature review; Information management

1. Introduction

A new type of passport has entered the agenda of sustainable supply chains and circular economy: the digital product passport (DPP). Its purpose is to store relevant product information in a reliable digital medium for circular practices and environmental impact that is valuable to the end users [26], which may influence their purchase intention [18]. Emerging from the European Green Deal, DPPs will be mandatory for products defined by the Ecodesign for Sustainable Products Regulation, and relevant European projects are paving the way to the DPP core [11]. Digital technologies will be essential to put this concept into practice, including data carriers, interoperability standards, or

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artificial intelligence (AI) to improve circular manufacturing. However, the literature on DPPs is still in the early stages of development, and the research opportunities are vast [17, 26], particularly in sustainable supply chain management.

Managing information within the entire product lifecycle is challenging. Two reasons are the particularities of each economy sector and the need for a transdisciplinary approach to obtain, store, process, and share product information relevant to sustainability. Therefore, recent studies report DPP proposals for different products, most notably batteries [8] and textiles [19, 40]. There are also design principles suggested in hydrogen supply chains [17], making it obvious that ecosystem cooperation and data from multiple stakeholders are necessary to create a DPP [2]. Nevertheless, many proposals "develop information frameworks using bottom-up approaches that structure the information into groups based on the information points they have identified (…) resulting in critical viewpoints and requirements being obfuscated or missed" [45]. There are insufficient studies addressing data governance [13] and an excellent opportunity to integrate research advancing in different fields (e.g., law, digital transformation) to understand the current landscape in DPP research and the avenues for future work in the field.

Several reviews were found on DPPs, justifying an analysis of related work and the existing gaps. Web of Science (WoS) identifies two excellent examples, one for DPPs in plastics [39] and the other published in 2021 [48], establishing the link with the circular economy, explaining the DPP concept, and including the specific case of batteries. Scopus allows the extension of the sample with a very recent paper [46], detailing technologies available for DPPs in the domains of data collection (e.g., Internet of Things (IoT) potential, QR code), curation and sharing, and data leverage, highlighting the role of blockchains. Due to the novelty of DPPs, multiple papers adopt literature review strategies as an essential dimension of the contribution. Environmental information about product transportation and logistics is key for DPPs. Still, we could not find a review providing a global overview of the DPP evolution since its proposal to the most recent boost of publications in 2023/2024, which leads to our overall research objective: *understand the research landscape on DPPs and make sense of key concepts necessary to put them in practice*.

The rest of the paper is structured as follows. Section 2 presents the approach used to revise the literature on DPPs. Subsequently, the results are presented, and section 4 highlights key findings and a research agenda. Section 5 summarizes the conclusions and study limitations.

2. Method

There are essential references available for systematic literature reviews. Our work follows the steps proposed by Okoli & Schabram [30], starting with clarifying the purpose, protocol, and training to ensure consistent analysis by two researchers. The search was conducted during February and March 2024 using the keyword "digital product passport" with no time restrictions. The team selected WoS and Scopus, two of the most prominent scientific databases. A total of 18 papers from WoS and 145 from Scopus, published in English, were screened in the title and abstract during the first stage of the review process. After removing duplicates and papers that did not focus on DPP (e.g., papers briefly mentioning the term; circular economy studies where DPPs had a secondary role), 40 papers remained in our sample for full-text analysis. The two researchers classified each paper using Excel and Mendeley reference management software and conducted a bibliometric analysis to identify key terms and the global landscape in this area using VoSViewer [14]. Subsequently, a concept-centric approach was selected for synthesizing and writing the review [49]. The research team collaborated during the entire process to confirm each decision (e.g., studies exclusion/inclusion) and make a continuous reflection on the findings.

Our approach focuses on (1) the anatomy of scientific research, (2) the envisioned cross-sectoral structure for DPPs, (3) enabling technologies, (4) requirements, (5) challenges, and (6) avenues for future work. The results are subsequently presented.

3. Results

3.1. Bibliometric analysis of DPP research

The first paper in our sample appeared in 2021, with an exceptional increase: 2 in 2021, 4 in 2022, 31 in 2023, and 3 in the first quarter of 2024. The number of conference papers (47%) and journal articles (53%) are balanced.

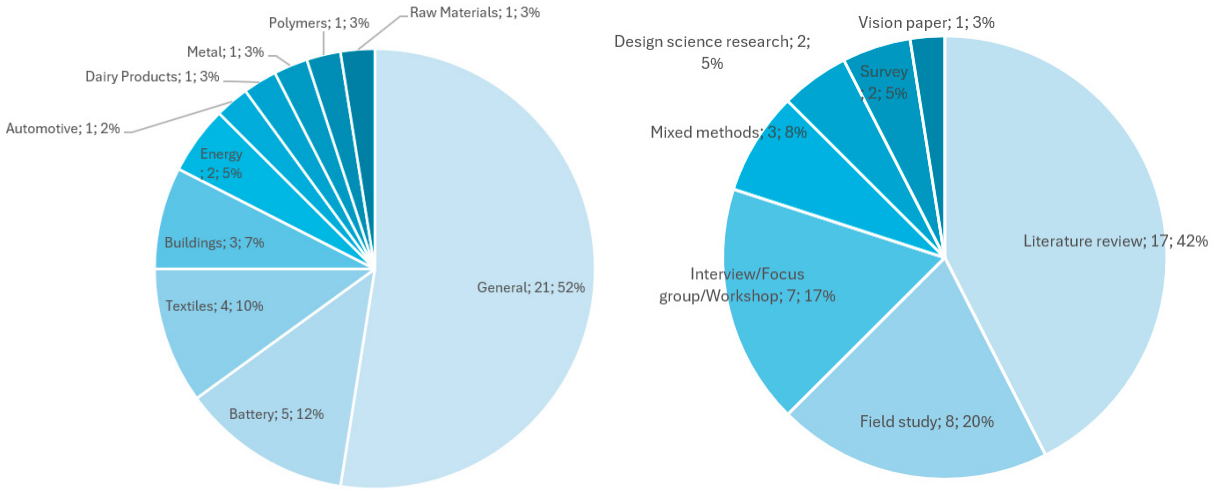


Fig. 1. (a) product type; (b) methods.

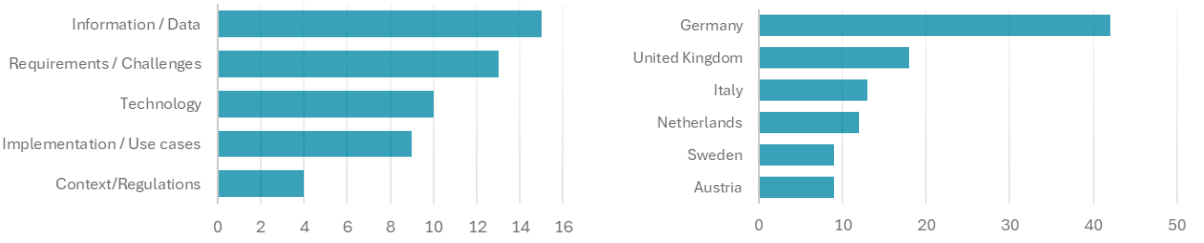


Fig. 2. (a) paper focus; (b) top countries (source: Scopus).

DPPs can be applied to different products, and as expected, priorities like batteries and textiles are well represented (Fig. 1a), with 5 and 4 papers, respectively. Interestingly, a significant share of the documents (52%) is not specific to a product. The term captured the attention of the construction, manufacturing (e.g., automotive), and materials sectors (e.g., polymers, metal, raw materials), but also food, which is consistent with the traceability concerns or zero km initiatives.

Aligned with the main goal to support the circular economy, most papers are included in the sustainable development goals 12 – Responsible Production and Consumption (prevalent) and 13 – Climate Action. Fig. 1b reveals that nearly half of the contributions are primarily conceptual (some works using mixed methods also rely significantly on literature reviews), but several examples address practitioners' perspectives and collaboration with companies interested in DPP research. The bibliometric analysis also identifies many papers focusing on the information that should be included in DPPs and data-related issues (e.g., privacy, security, trust), requirements and challenges faced by companies, and the technology line of research in the third position (Fig. 2a). Some studies also present the development of DPP in specific sectors and aspects related to the adoption context, like regulations. Fig. 2b reveals that some of the most advanced economies in Europe (especially in manufacturing) are the most interested in DPP research, with Germany leading by a large margin.

Fig. 3 presents the keyword analysis obtained with the Scopus sample, using a thesaurus file to exclude meaningless terms and associate similar concepts.

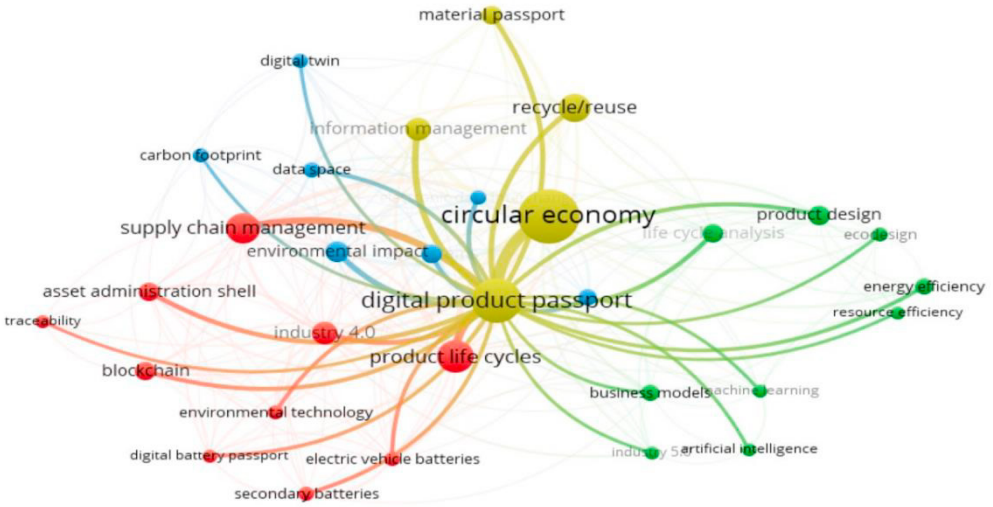


Fig. 3. Relevant keywords in Scopus sample using a bibliometric network.

The analysis reveals technological trends like blockchain, which is included in the red cluster of batteries (on the left), and traceability concerns. The digital twin (a digital replica of physical objects or systems with monitoring and predictive capabilities) also looks promising in addressing environmental impacts. The green cluster includes more recent publications in Industry 5.0 and the importance of artificial intelligence to ensure efficiency and eco-design. The close association with material passports can be found in the yellow cluster – association between DPPs, circular economy, information management, and the need to reuse/recycle towards circular practices.

3.2. Concept-centric analysis

This section details four key concepts: DPP structure, technology, requirements, and challenges.

3.2.1. DPP structure

Table 1 identifies the most common structural elements of a DPP listed in the literature.

Table 1. DPP information.

Information	Details/Examples	Reference
Product Information	Materials sourcing data (Date and Place); Compositional elements (materials, substances, components & parts); Properties (physical and chemical properties, material criticality, bio-sustainability, recycled content, reused/refurbished parts, hazardous content to human health or the environment)	[5, 22, 32, 42, 45]
Environmental Information	Environmental footprint; Carbon footprint; Reusing, Reprocessing, Recycling Data; Information about the product in terms of sustainability and circularity properties (e.g., environmental impact, social impact, circularity performances, disassembly instructions, inventory data used for calculation, calculation methods, indicators for environmental/social/circularity performance, applied standards for impact assessment, product design)	[5, 22, 32, 42, 45, 47]
Manufacturing Data	Manufacturing process and remanufacturing, Resource consumption/Consumption in production (water, energy, etc.)	[16, 22, 32, 42, 45, 47]
Utilization Information	Documentation of any parts of a product that have been replaced or repaired, maintenance history, delivered performance	[5, 22, 32, 42, 45]

Information	Details/Examples	Reference
Life-cycle Data	Expected percentages of products that can be recycled and the waste part; Documentation on collection, sorting, and treatment during the end-of-life of a product; Recovery Data	[5, 22, 45]
Supply chain	Suppliers of components; Recycling partners; Customer return channels; Implementation of circular economy business models; Information about value chain actors involved	[4, 5, 22, 32, 42, 45]
Guidelines and Manuals	Product manuals; Waste sorting guidelines; Safety instructions	[22, 45]
Compliance	Regulatory/voluntary compliance; Product-specific regulation; Labor conditions	[22, 42, 45]

Regardless of the type of product (generic, batteries, or energy), the most usual information suggested for DPPs is related to the product attributes, the process, and materials used in manufacturing, product use, and environmental information. Several authors highlight the product's value chain and useful life as essential elements to be included in the DPP. Life-cycle data, supply chain, compliance, guidelines, and manuals are also essential.

3.2.2. DPP technology

Fig. 4 highlights the most popular technologies mentioned in DPP research.

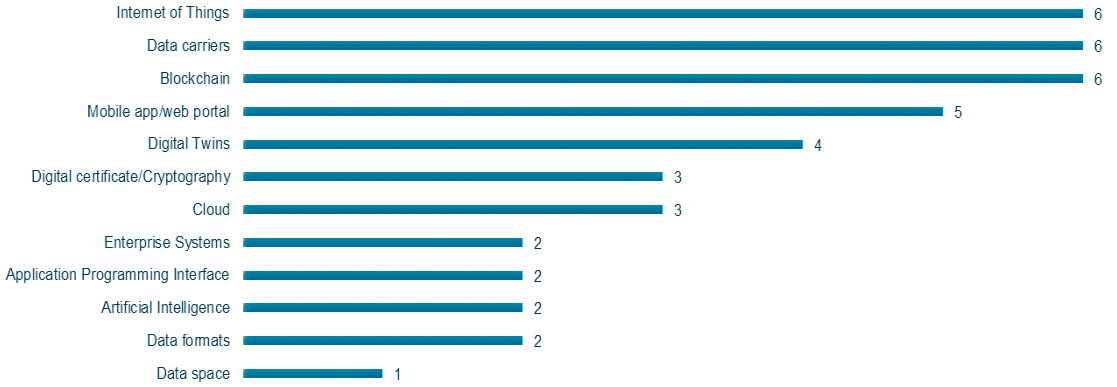


Fig. 4. DPP technologies.

Adopting sensing technologies in different stages of the supply chain (e.g., GPS for product location, process parameters monitoring), data carriers (e.g., bar code, QR code, RFID), and blockchain are leading the ranking of most popular technologies for DPP implementations. Five authors addressed how to visualize product information with mobile applications or portals [34], followed by the digital twin. A digital twin aims to reproduce a physical object/system with real-time monitoring and decision-support capabilities. Therefore, it includes other technologies in a more complex solution and is one of the most promising approaches to incorporate product data along the lifecycle while also assisting manufacturing and logistics in actions that can reduce impacts (e.g., predictive maintenance).

Data storage (off-chain, using cloud platforms, which can also be used in parallel with blockchain for more critical information) and protection using cryptography techniques will also be essential to provide trust to consumers. Less mentioned technologies but equally relevant include enterprise systems (e.g., ERP integration), APIs, artificial intelligence (not yet sufficiently explored in the context of DPPs), and standards for data exchange (e.g., XML). One of the papers explores the creation of data spaces that can be relevant to share data across different stages of the product lifecycle [36]. The shortcoming in this sample is that most proposals are only conceptual, two are prototypes, and only one project in textile reveals a higher technology readiness level (TRL).

3.2.3. DPP Requirements

Seventeen main types of requirements are presented in Fig. 5.

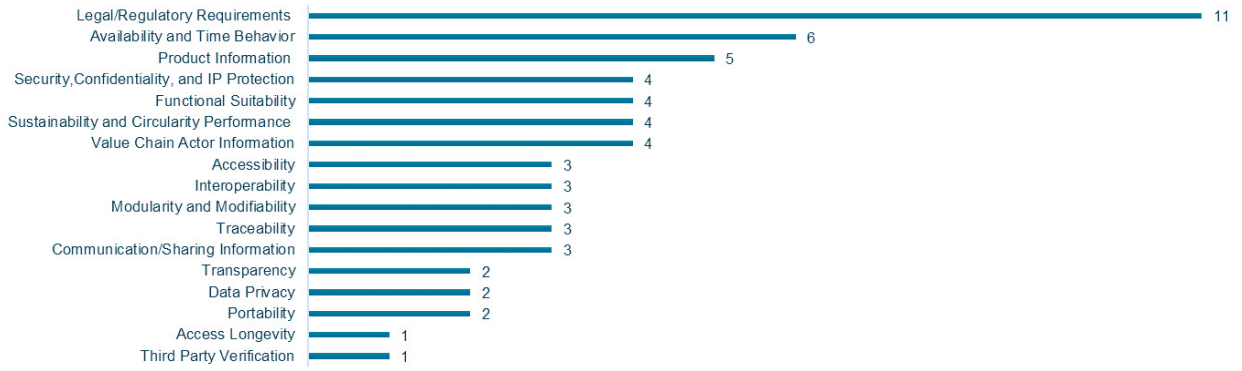


Fig. 5. DPP requirements.

The need for a solid regulatory framework is particularly relevant and can be identified in 11 studies [2], [45], [39], [48], [22], [32], [5], [38], [20], [24], [35], followed by the real-time availability of product information [32], [38], [20], [24], [35], [10]. Requirements for the information included in DPP are analyzed in five works [39], [5], [24], [35], [10]. Product sustainability, circularity performance [2], [5], [24], [35], functional suitability [2], [32], [38], [20], and value chain actor information [45], [22], [5], [35] are of the utmost importance to consumers while maintaining security, confidentiality, and intellectual property (IP) protection [39], [22], [32], [20]. Data requirements include accessibility [32], [20], [24], [10], interoperability [32], [20], [24], modularity and modifiability (e.g., change in one component has minimal impact on another) [32], [38], [20], traceability [5], [24], [35] and communication/sharing information [24], [35], [15]. Finally, the requirements of transparency [5], [31], data privacy [32], [20], portability [32], [20], access longevity [24], and third-party verification [24] were also mentioned.

These requirements address interrelated dimensions of (1) mandatory/voluntary regulations (e.g., privacy, IP protection, verification), (2) data since the early stages of data identification and collection, preferably with IoT, including sharing and retention over time, (3) business-related (e.g., transparency, value chain iterations, and availability), and (4) technical (e.g., cybersecurity, modules, interoperability standards). However, they seem to uncover only the top of the iceberg when we consider the voluntary disclosure of information and use of the DPP, for example, to provide valuable data for AI approaches, namely, supplier product selection based on DPP data and final product requirements, benchmarking and alternative product classification, or prediction of combined circularity potential within complex supply chains. The next section summarizes the challenges in the four main types of requirements.

3.2.4. Challenges

Table 2 summarizes the findings.

Table 2. DPP challenges.

Scope	Challenge	Reference
Regulations	Balance data access with privacy requirements	[47],[13][17],[26],[38]
	Standards compatibility/overlapping or differences across sectors/regions; Harmonization	[28],[6],[26],[9],[27]
	Compliance costs	[47]
	Deal with cross-sectoral differences/stakeholders' interests / unique DPP structure	[47],[29],[47],[10],[22]
Data	Trustworthy data exchange	[13],[2],[7],[17],[43],[26],[20]
	The data lifecycle and physical product lifecycle may not match. For example, data about raw materials or data collected during product use and end-of-life	[13],[47],[26],[47],[33]
	Data processing necessary to produce the DPP	[13],[2],[7] [15],[17],[26],[22],[20],[33],[38]

Scope	Challenge	Reference
	The granularity of data collection for CO2 emissions / Process vs product level	[2],[15],[47]
	Data intermediaries	[13]
	Data persistence over long periods of time	[15],[20]
Business	Supply chain negotiation and collaboration/data sharing incentives	[2],[7],[15],[6],[38]
	Different velocities and capabilities when comparing large companies and SMEs	[19]
	Green energy is still developing in some sectors	[2],[17]
	Regulatory pressures to report sustainable practices	[2]
	Transition period with products already available in the market and new products	[25]
	Transparency may reduce negotiation margins	[2]
	Comparison with competitors	[2],[26]
	Additional costs to implement DPP and the data infrastructure necessary in the company and in other supply chain elements	[2],[28],[10],[38]
Technical	Accessibility of the DPP by humans and machines	[20],[37]
	Different stakeholders' permissions in the DPP creation	[37]
	Product tags can be altered	[15]
	System infrastructure robustness and resiliency, dealing with large volumes of data	[17],[47],[6]

Solutions (some addressing more than one challenge) can also be found. For example, there is a need for a data governance model, enforcement and monitoring activities in the supply chain and data processing activities, data anonymization, adopting more sensors to obtain real-time product information, and using blockchain. However, blockchain is not mandatory, and there are challenges to using blockchain in such a distributed network (e.g., inter-blockchain issues, costs, energy resources, blockchain operator's uncertainties) that need additional research.

4. Discussion and Outlook

DPPs are being introduced internationally, influenced by circular economy priorities [41], and a new vision for Industry 5.0, human-centric, sustainable, and resilient. In December 2023, the European Parliament voted on the final text of the Proposal for a Regulation of the European Parliament and of the Council establishing a framework for defining ecodesign requirements for sustainable products in which it determines DPP vision, which is expected to complement the existing product labels [12].

The DPP will significantly impact the logistics sector, including seaport operations, dry ports, and intermodal terminals. On the one hand, it is justified by the need to include supply chain information. On the other hand, the maritime industry already included DPPs in their research agenda for the circular economy [21], and major companies in this area, like Maersk, have created their vision for passports, for example, with the C2C (cradle-to-cradle) passport for recycling/reusing steel, revealing the importance of this type of documents in logistics [24]. Our study may assist companies operating in the logistics field in preparing for the requirements and also taking the lead in DPP research.

Implementing the DPP requires tracking the raw materials and product components from diverse geographic regions, and asymmetries may appear. First, the coexistence of environmental regulations specific to different parts of the globe. Second, information asymmetry between manufacturers, intermediaries, and consumers may create risks for product information trustworthiness. Third, there are possible inequalities between SMEs and large companies in the capacity to obtain, process, and disclose required information within the entire product lifecycle.

The "DPP artifact" structure will need information from multiple stakeholders. Examples include end-users (*how DPPs influence purchasing decisions?; how can DPP be used to propagate the users' feedback to different supply chain segments?*), policymakers (*how to maximize DPP value? how to balance the cost-benefit of mandatory DPP data?*), supply chain participants (*how DPPs can be decoupled in the product components and supply chain stages? how can interoperability and compliance in DPP creation be ensured? how to manage DPP data and workflow? how to differentiate business with DPPs*). Table 3 summarizes essential research opportunities for future DPP research.

Table 3. An annotated bibliography for future research in DPP.

Scope	Opportunities	Reference
DPP context	Research how DPPs can be part of an integrated approach to sustainable product management and the priorities for product adoption	[8],[28],[27],[1]
	Integrate DPPs in Industry 5.0 research and policies under development	[26],[46],[24][27],[1]
	Cross-sectoral studies to identify DPP synergies, common problems, and possible solutions	[2],[22]
	DPP standardization (and integration with other standards) and data	[26],[48],[37]
DPP adoption	Identify drivers, enablers, and barriers to DPP adoption in different sectors of the economy	[8],[22]
	Study how DPPs can be generated with minimum effort and resources, for example, taking advantage of usual data collection moments (e.g., maintenance procedures)	[32],[20],[10],[1]
	Digital business ecosystems, DPP interoperability, and inter-organizational data governance	[17],[22],[42],[24]
	Empirical/case studies in company settings or supply chain settings	[17],[23]
DPP content	The potential of machine learning, open data, and data science and engineering for DPP data	[46],[24],[10],[7]
	Detail the data required for each DPP according to the context of use (and the associated rationale and cost-benefit) and granularity – company level, product or component, process	[2],[48],[42],[15]
	Validate AI techniques and the incorporation of users' feedback or crowdsourcing potentiality	[32],[47],[10],[9]
	Studies in different countries, product typologies, and supply chains to identify requirements	[2],[22],[10]
	Understand the integration of DPPs with other types of passports (e.g., material passports) and frameworks or proposals like the Asset Administration Shell	[17],[1]
	Address the challenges of data privacy, security, storage, and sharing, including the potential of cryptography and the requirements of intellectual property	[48][46],[20],[10],[15],[29]
DPP technology	Adoption of blockchain for DPP infrastructure	[46],[22],[10],[29]
	IoT developments to provide data in real-time for DPPs, in each step of the product lifecycle	[26],[46],[29]
	Digital twin implementations that mirror the physical product and its supply chain	[48],[46]
DPP use	Data accessibility and availability	[8],[48],[42],[10]
	Link the role of passports and the data points necessary to create them	[20],[10],[3]
	Study how DPPs can have a preventive/predictive role to anticipate and/or prevent sustainability problems in the supply chain	[48],[24],[28],[27]
	Relevance to companies and possible incentives, not only as a compliance requirement	[24],[15],[28]
	User experience studies are necessary to improve DPP interaction and the end users' interest	[37],[1]
Impact & Outcomes	Research the economic and environmental impacts of DPPs	[38],[43]
	Regulatory implications of DPP and liability issues	[26],[9]

The opportunities in Table 3 are sociotechnical and provide a clearer picture of the enormous research potential in DPPs. Interestingly, most of the opportunities can be explored immediately. For example, technologies supporting DPPs, research on data-related issues like security, privacy, or analytics, and DPP content. Multiple studies point to the cross-sectoral / ecosystem nature of DPPs and the close link with Industry 5.0 efforts at the European level. DPP should not be a burden, requiring studies about incentives for companies, integration in existing information systems, crucial data entry points necessary for different stakeholders, generating value, and, simultaneously, being sustainable in its adoption, minimizing energy and resources.

5. Conclusion

This paper presented a systematic literature review and a bibliometric analysis to understand the current research landscape on DPP. A total of 40 publications were chosen from WoS and Scopus and evaluated in the domains of (1) the anatomy of scientific research, (2) the envisioned cross-sectoral structure for DPPs, (3) enabling technologies, (4) requirements, (5) challenges, and (6) avenues for future work. The DPP will change supply chain information flows.

Nevertheless, several limitations must also be stated. First, this article focused on information about the "DPP artifact", not detailing the sources of information provided by the companies operating in the product's life cycle. These stakeholders are responsible for adding information to the DPP (e.g., in product development or transportation), forming a network of complex interactions that may vary in each sector of the economy. Second, we selected important databases to support our study and a specific keyword, but other databases can also be used. Grey literature can provide additional insights, particularly in illustrating practical implementations. Third, DPPs are still under development in European institutions, and most of the work is conceptual in nature. Future research in DPP must consider cooperation with users to improve its usability and enrich the available information, following the paradigm of open innovation theory [44], and currently under discussion for human-centric and sustainable product redesign. We hope this study can provide a starting point for additional studies in this new complement for product labels with significant implications for circular logistics, responsible production, and responsible consumption.

Acknowledgments

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References

[1] Adisorn, Thomas, Lena Tholen, and Thomas Götz. (2021) "Towards a Digital Product Passport Fit for Contributing to a Circular Economy". *Energies* **14**:2289.
[2] Alt, Shari, Victoria Schorr, Anna Vocke, and Dirk Werth. (2023) "Status and Future of Real-Life Application of the Digital Process Passport in Germany". *Circular Economy and Sustainability* **3**:2255 – 2266.
[3] Alves, Luís, Miguel Sá, Estrela Ferreira Cruz, Toni Alves, Marcelo Alves, João Oliveira, Manuel Santos, and António Miguel da Cruz. (2024) "A Traceability Platform for Monitoring Environmental and Social Sustainability in the Textile and Clothing Value Chain: Towards a Digital Passport for Textiles and Clothing". *Sustainability* **16**:82**.**
[4] Bar, Martin Alejandro, Armando Walter Colombo, Jose Luis Torres, Erica Fernandez, Mariela Rico, and Maria Laura Caliusco. (2023) "An Industry 4.0-Compliant Digital Product Passport Approach for Realising Dairy Product Traceability". In: IECON Proceedings.
[5] Berger, Katharina, Rupert J Baumgartner, Martin Weinzerl, Johann Bachler, Kees Preston, and Josef-Peter Schöggl. (2023) "Data requirements and availabilities for a digital battery passport – A value chain actor perspective". *Cleaner Production Letters* **4**:100032**.**
[6] Berger, Katharina, Rupert J Baumgartner, Martin Weinzerl, Johann Bachler, and Josef-Peter Schöggl. (2023) "Factors of digital product passport adoption to enable circular information flows along the battery value chain". In: Procedia CIRP. pp 528 – 533.
[7] Berger, Katharina, Magdalena Rusch, Antonia Pohlmann, Martin Popowicz, Bernhard C Geiger, Heimo Gursch, Josef-Peter Schöggl, and Rupert J Baumgartner. (2023) "Confidentiality-preserving data exchange to enable sustainable product management via digital product passports - a conceptualization". In: Procedia CIRP. pp 354 – 359.
[8] Berger, Katharina, Josef-Peter Schöggl, and Rupert J Baumgartner. (2022) "Digital battery passports to enable circular and sustainable value chains: Conceptualization and use cases". *Journal of Cleaner Production* **353**:131492.
[9] De Bruyne, Jan, Orian Dheu, and Charlotte Ducuing. (2023) "The European Commission's approach to extra-contractual liability and AI – An evaluation of the AI liability directive and the revised product liability directive". *Computer Law and Security Review* **51**:105894.
[10] Çetin, Sultan, Deepika Raghu, Meliha Honic, Ad Straub, and V. Gruis. (2023) "Data requirements and availabilities for material passports: A digitally enabled framework for improving the circularity of existing buildings". *Sustainable Production and Consumption* **40**:422 – 437.
[11] CIRPASS. (2023) "Digital Product Passport". https://cirpassproject.eu/.
[12] Council of the European Union. (2023) "Regulation establishing a framework for setting ecodesign requirements for sustainable products and repealing Directive 2009/125/EC - Analysis of the final compromise text for an agreement".
[13] Ducuing, Charlotte, and René Herbert Reich. (2023) "Data governance: Digital product passports as a case study". *Competition and Regulation in Network Industries* **24**:3 – 23.
[14] van Eck, Nees Jan, and Ludo Waltman. (2010) "Software survey: VOSviewer, a computer program for bibliometric mapping". *Scientometrics* **84**:523–538.
[15] Van Engelenburg, Selinde, Boriana Rukanova, Jolien Ubacht, Siu Lie Tan, Yao Hua Tan, and Marijn Janssen. (2022) "From requirements to a research agenda for governments governing reuse of critical raw materials in the circular economy". In: 23rd DGO. pp 62 – 67.
[16] Gallina, Viola, Barna Gal, Ádám Szaller, Daniel Bachlechner, Elisabeth Ilie-Zudor, and Wilfried Sihn. (2023) "Reducing Remanufacturing Uncertainties with the Digital Product Passport". *Lecture Notes in Mechanical Engineering* 60 – 67.
[17] Heeß, Paula, Jakob Rockstuhl, Marc-Fabian Körner, and Jens Strüker. (2024) "Enhancing trust in global supply chains: Conceptualizing Digital Product Passports for a low-carbon hydrogen market". *Electronic Markets* **34**:10.
[18] Hinnüber, Felix, Marek Szarucki, and Katarzyna Szopik-Depczyńska. (2019) "The Effects of a First-Time Experience on the Evaluation of Battery Electric Vehicles by Potential Consumers". *Sustainability* **11**:7034.

[19] Jæger, Bjørn, and Sivert Myrold. (2023) "Textile Industry Circular Supply Chains and Digital Product Passports. Two Case Studies". *IFIP Advances in Information and Communication Technology* **692**:350 – 363.
[20] Jansen, Maike, Tobias Meisen, Christiane Plociennik, Holger Berg, André Pomp, and Waldemar Windholz. (2023) "Stop Guessing in the Dark: Identified Requirements for Digital Product Passport Systems". *Systems* **11**:123.
[21] Jensen, Henrik Hvid, Henrik Sornn-Friese, Steffen Foldager Jensen, and Nicolò Aurisano. (2023) "The Implications of Circular Supply Chains and the EU Digital Product Passport in Maritime Decarbonization". In: Maritime Decarbonization. Springer, pp 231–250.
[22] Jensen, Steffen Foldager, Jesper Kristensen, Sofie Adamsen, Andreas Christensen, and Brian Waehrens. (2023) "Digital product passports for a circular economy: Data needs for product life cycle decision-making". *Sustainable Production and Consumption* **37**:242 – 255.
[23] Kebede, Rahel, Annika Moscati, He Tan, and Peter Johansson. (2023) "Circular economy in the built environment: a framework for implementing digital product passports with knowledge graphs". In: Proceedings of the European Conference on Computing in Construction.
[24] King, Melanie RN, Paul D Timms, and Sara Mountney. (2023) "A proposed universal definition of a Digital Product Passport Ecosystem (DPPE): Worldviews, discrete capabilities, stakeholder requirements and concerns". *Journal of Cleaner Production* **384**:135538.
[25] Koppelaar, Rembrandt HEM, Sreenivaasa Pamidi, Enikő Hajósi, Lucia Herreras, Pascal Leroy, Ha Young Jung, Amba Concheso, Radha Daniel, Fernando B Francisco, Cristina Parrado, Siro Dell'Ambrogio, Fabiana Guggiari, Deborah Leone, and Alessandro Fontana. (2023) "A Digital Product Passport for Critical Raw Materials Reuse and Recycling". *Sustainability* **15**:1405**.**
[26] Langley, David J, Eugenia Rosco, Marios Angelopoulos, Oscar Kamminga, and Christa Hooijer. (2023) "Orchestrating a smart circular economy: Guiding principles for digital product passports". *Journal of Business Research* **169**:114259.
[27] Lövdahl, Josefin, Sophie I Hallstedt, and Jesko Schulte. (2023) "Implications of EU instruments on company capabilities to design more sustainable solutions–Product Environmental Footprint and Digital Product Passport". *Proceedings of the Design Society* **3**:2245–2254
[28] Mêda, Pedro, Mustapha Munir, Diego Calvetti, and Hipólito Sousa. (2023) "Information needs in digital products passport – discussing data framework and reasonability". In: Proceedings of the European Conference on Computing in Construction.
[29] Nowacki, Szymon, Gokay Meric Sisik, and Constantinos Marios Angelopoulos. (2023) "Digital Product Passports: Use Cases Framework and Technical Architecture Using DLT and Smart Contracts". In: DCOSS-IoT 2023. pp 373 – 380.
[30] Okoli, Chitu, and Kira Schabram. (2010) "A Guide to Conducting a Systematic Literature Review of Information Systems Research". *Sprouts Working Papers on Information Systems* **10**:1–49.
[31] Ospital, Pantxika, Dimitri Masson, Cédrick Beler, and Jérémy Legardeur. (2023) "Toward product transparency: communicating traceability information to consumers". *International Journal of Fashion Design, Technology and Education* **16**:186–197.
[32] Panza, Luigi, Giulia Bruno, and Franco Lombardi. (2023) "Integrating Absolute Sustainability and Social Sustainability in the Digital Product Passport to Promote Industry 5.0". *Sustainability* **15**:12552.
[33] Piétron, Dominik, Philipp Staab, and Florian Hofmann. (2023) "Digital circular ecosystems: A data governance approach". *GAIA - Ecological Perspectives for Science and Society* **32**:40 – 46.
[34] Plociennik, Christiane, Monireh Pourjafarian, Ali Nazeri, Waldemar Windholz, Svenja Knetsch, Julian Rickert, Andreas Ciroth, Alice Do Carmo Precci Lopes, Tabea Hagedorn, Malte Vogelgesang, W. Benner, Andrea Gassmann, Simon Bergweiler, Martin Ruskowski, Liselotte Schebek, and Anke Weidenkaff. (2022) "Towards a Digital Lifecycle Passport for the Circular Economy". In: Procedia CIRP. pp 122 – 127.
[35] Plociennik, Christiane, Monireh Pourjafarian, Shehab Saleh, Tabea Hagedorn, Alice do Carmo Precci Lopes, Malte Vogelgesang, Julian Baehr, Bernd Kellerer, Maike Jansen, Holger Berg, Martin Ruskowski, Liselotte Schebek, and Andreas Ciroth. (2022) "Requirements for a Digital Product Passport to Boost the Circular Economy". In: Lecture Notes in Informatics. pp 1485 – 1494.
[36] Pomp, André, Maike Jansen, Holger Berg, and Tobias Meisen. (2023) "SPACE_DS: Towards a Circular Economy Data Space". In: Companion Proceedings of the ACM Web Conference 2023. pp 1500–1501.
[37] Pourjafarian, Monireh, Christiane Plociennik, Mohammad Hossein Rimaz, Peter Stein, Malte Vogelgesang, Chanchan Li, Svenja Knetsch, Simon Bergweiler, and Martin Ruskowski. (2023) "A Multi-Stakeholder Digital Product Passport Based on the Asset Administration Shell". In: IEEE International Conference on Emerging Technologies and Factory Automation, ETFA.
[38] Psarommatis, Foivos, and Gökan May. (2024) "Digital Product Passport: A Pathway to Circularity and Sustainability in Modern Manufacturing". *Sustainability* **16**:396.
[39] Rumetshofer, Thomas, and Jörg Fischer. (2023) "Information-Based Plastic Material Tracking for Circular Economy—A Review". *Polymers* **15**:1623.
[40] Sá, Miguel, Catarina Guise, Filipa Costa, Paula Rodrigues, João Oliveira, Ana Barros, Ricardo Silva, Toni Alves, Manuel Santos, António Miguel Rosado da Cruz, Marcelo Alves, and Carla Joana Silva. (2024) "STVgoDigital: A Digital Product Passport Solution". *Communications in Computer and Information Science* **1937**:368 – 376.
[41] Stahel, Walter R. (2016) "The circular economy". *Nature* **531**:435–438.
[42] Stratmann, Lukas, Gerrit Hoeborn, Christoph Pahl, and Günther Schuh. (2023) "Classification of Product Data for a Digital Product Passport in the Manufacturing Industry". In: Proceedings of the Conference on Production Systems and Logistics. pp 448 – 458.
[43] Szaller, Ádám, Viola Gallina, Barna Gal, Alexander Gaal, and Christian Fries. (2023) "Quantitative benefits of the digital product passport and data sharing in remanufacturing". In: Procedia CIRP. pp 928 – 933.
[44] Szopik-Depczyńska, Katarzyna, Katarzyna Cheba, Iwona Bąk, and Giuseppe Ioppolo. (2019) "User-Driven Innovation in Poland: Determinants and Recommendations". *Sustainability* **12**:171.
[45] Timms, Paul D, and Melanie RN King. (2023) "Complexity in the delivery of product passports: a system of systems approach to passport lifecycles". In: 2023 18th Annual System of Systems Engineering Conference (SoSe). IEEE, pp 1–8.
[46] Voulgaridis, Konstantinos, Thomas Lagkas, Constantinos Marios Angelopoulos, Alexandros-Apostolos A Boulogeorgos, Vasileios Argyriou, and Panagiotis Sarigiannidis. (2024) "Digital product passports as enablers of digital circular economy: a framework based on technological perspective". *Telecommunication Systems* **85**:699–715.
[47] Voulgaridis, Konstantinos, Thomas Lagkas, Dimitris Karampatzakis, Vasileios Argyriou, and Panagiotis Sarigiannidis. (2023) "Realizing Digital Product Passports with Crowdsourcing Principles: The Case of Sustainable Smart Grids". In: DCOSS-IoT 2023. pp 381 – 388.
[48] Walden, Joerg, Angelika Steinbrecher, and Maroye Marinkovic. (2021) "Digital Product Passports as Enabler of the Circular Economy". *Chemie Ingenieur Technik* **93**:1717–1727.
[49] Webster, J, and RT Watson. (2002) "Analyzing the past to prepare the future". *MIS Quarterly* **26**:xiii–xxiii.